Premises to the Project Submission

Francesco Bassignana

1 Use of AI Tools

I used AI assistance to help generate code and documentation for this project; the code, its extended documentation and the experiments were reviewed and run by me. I also made use of AI tools in writing the report.

2 Scope of the Report

As discussed with the supervisor in the last meeting, before working with the MusicGen model I explored masked-autoencoder (MAE) and diffusion-based solutions. Their results were qualitatively poor early on, so I chose not to pursue them further, and they are therefore not included in the report.

3 Selection of the Demonstration Samples

The demonstration samples let a listener compare, on identical gaps, the audio produced by the trained models against the original and against leaving the gap silent. They are meant to illustrate the findings of the report, not to present the models more favourably than the metrics do. Gaps are 13–53 ms long in solo-piano music, and every fill is spliced into the original so that only the samples inside the gap differ.

Models and checkpoints. Each gap is rendered by the following versions: the original audio; the zeroed input (the "do nothing" reference); the Phase 6 final run (`phase6_final_songs17_seed43`, tag `final`, 300k steps); the two matched 17-song seeds (`phase3_songs17_seed43` and `seed44`, tag `final`, 300k steps); the long 17-song run (`phase1_songs17`, tag `best_val`, step 950k); and the single-song run (`phase1_songs01`, tag `final`). The early `best_val` checkpoints (Phase 6 at step 30k, songs17 seed 43 at step 20k) are excluded, since they correspond to validation minima reached long before the models are trained.

Real gaps. The four real gaps of 15, 20, 30 and 50 ms are included without selection, since they are the use case the work targets. They are filled with the repository's own inference path (`Trainer.inpaint_all_gaps`), with splice output for every model.

Held-out gaps. Six to eight further gaps come from the 128-gap held-out set of `phase_6/eval_final.py`, using its gap sampling and trainer reconstruction. Within each of the three length bands we take the *best*, the *median* and the *worst* gap, ranked by gap-only SI-SDR averaged over all models. Showing the extremes together with the median avoids presenting only favourable cases. Filling follows the common protocol of that evaluation: the whole song is re-encoded with a real-like zeroed span, in fp32, with greedy codebook decoding, and the gap samples are spliced back with a 2.5 ms raised-cosine cross-fade.

Presentation. Each file is a 24 kHz mono crop of the gap ±1 s (about 2 s), loudness-matched across the versions of the same gap. An index lists, per file, the gap length, model, run, checkpoint, step, and gap-only and region SI-SDR.

4 Caveats

At these gap lengths, no 300k-step model beats the zeroed input on region SI-SDR under the common protocol, and the Phase 6 final run is 5.6 ± 1.8 dB worse than the two 17-song seeds on gap-only SI-SDR; the samples document this behaviour and do not conceal it.

5 Further Details

Because of the page limit, the report cannot contain all implementation details. For any question on the implementation, please refer to the codebase, which includes extended documentation of its structure (see `docs/SUPERVISOR_GUIDE.md`) as well as the code itself. The supervisor is of course also welcome to ask me any further question.